

Program : Diploma in Printing Technology	
Course Code : 5109B	Course Title: Troubleshooting Workshop
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3	Periods per semester: 45

Course Objectives:

- This course is designed to make the students aware about the critical and variable factors of the conventional offset printing method.
- Identifying the problems encounter during printing process and to find out it remedies.
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Course Prerequisites:

Topic	Course code	Course name	Semester
Inking System, Dampening System, Printing Unit.		Advanced Printing Machines Workshop I	3
Mechanism of Inking System, Dampening System, Printing Unit.		Mechanism of Printing Machines-I	3
Ink, Dampening solution, Plate, Blanket. Ink & Paper Problems.		Materials for Printing and Packaging.	3

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Identifying the troubles related to Paper and find out the remedies.	9	Applying
CO2	Identifying the troubles related to Ink and Dampening solution and find out their remedies.	10	Applying
CO3	Identifying the troubles related to Plate& Blanket and find out their remedies.	12	Applying
CO4	Identifying the troubles related to Printing machine and find out their remedies.	10	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	2		3	
CO2	3	3	3	2		3	
CO3	3	3	3	2		3	
CO4	3	3	3	2		3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identifying the troubles related to Paper and find out the remedies.		
M 1.01	Detect the problems related to paper in the offset printing process, identify the causes, and find out the solutions.	9	Applying
CO 2	Identifying the troubles related to Ink and Dampening solution and find out their remedies.		
M 1.01	Detect the problems occur to related to Ink, identify the causes, find out the solutions.	5	Applying
M 1.02	Detect the problems occurs related to dampening solution, identify the causes, find out the solutions.	5	Applying
	Lab exam - I	2	
CO 3	Identifying the troubles related to Plate& Blanket and find out their remedies.		
M 1.01	Detect the problems occurs related to lithographic printing plate, identify the causes, find out the solutions.	6	Applying
M 1.02	Detect the problems occurs related to offset blanket, identify the causes, find out the solutions.	6	Applying
CO 4	Identifying the troubles related to Printing machine and find out their remedies.		
M 1.01	Detect the problems occurs related to offset printing machine, identify the causes, find out the solutions.	10	Applying
	Lab Exam II	2	

Text / Reference:

T/R	Book Title/Author
T1	TulikaDas, Chemistry in Printing, 2nd Edition , BarnanaPrakashani, 2011.
R1	P. Kipphan, Handbook of Print Media , Springer, 2002.
R2	Bishwanath Chakravarty, A Hand Book for Printing and Packaging , Galgotia Publications Pvt. Ltd, 1997.
R3	GATF Staff Solving Offset Ink Problems , GATF, 1998
R4	N.R.Elred& T. Scarlet, What the Printer should know about Ink , GATF, 1995
R5	Herschel L Apfelberg, Maintaining printing equipment , GATF.
R6	HP Garg, Industrial maintenance , S Chand and company Ltd.

Online Resources:

Sl.No	Website Link
1	https://www.printwiki.org/Gravure
2	https://www.britannica.com/technology/flexography
3	https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-complete-beginner-c5bd393d9238
4	https://www.britannica.com/technology/offset-printing
5	http://edelmann-printingmachines.com